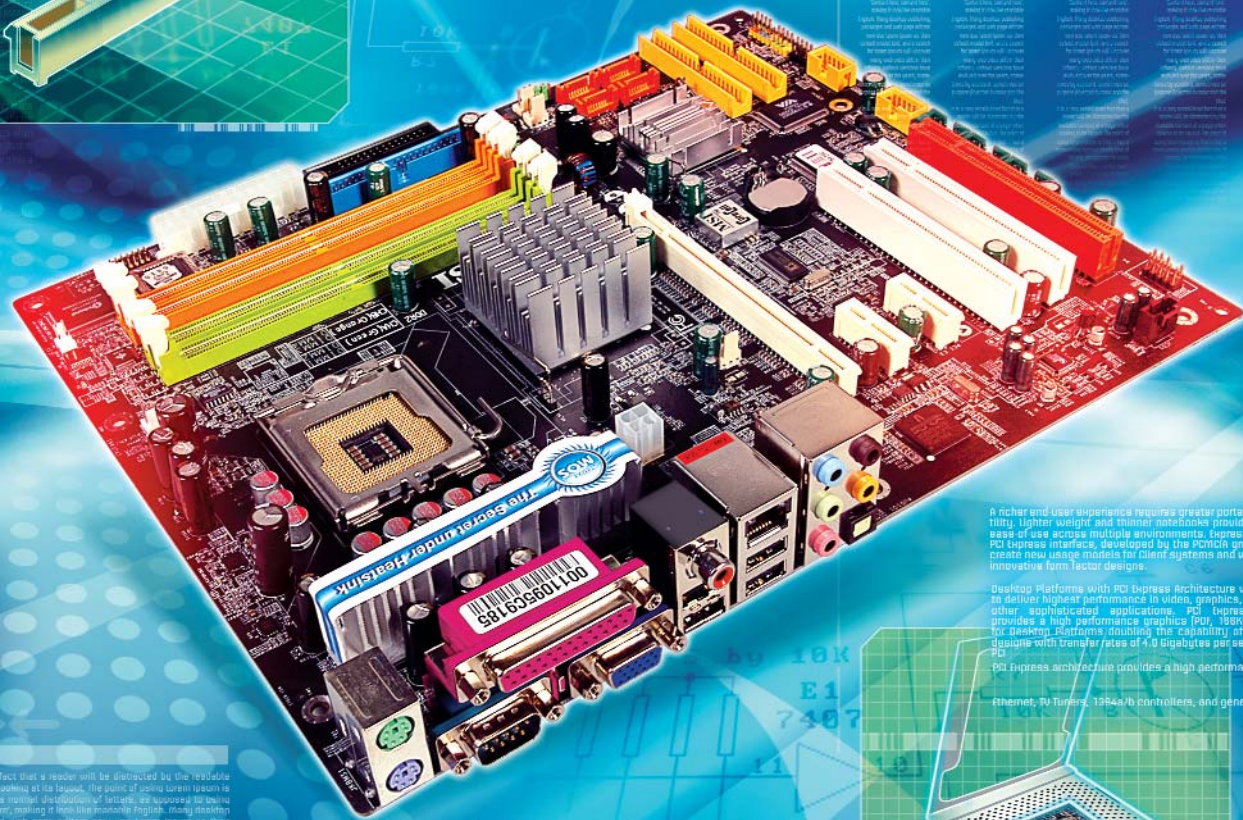




A richer and user experience requires greater portability and versatility. Lighter weight and thinner notebooks provide flexibility and ease-of-use across multiple environments. ExpressCard™ utilizing PCI Express interface, developed by the PCMCIA group, will help to create new usage models for Client systems and will demonstrate innovative form factor designs.

Desktop Platforms with PCI Express Architecture will be designed to deliver highest performance in video, graphics, multimedia and other sophisticated applications. PCI Express architecture provides a high performance graphics (PCI Express) infrastructure for desktop Platforms doubling the capability of existing x86 desktop with transfer rates of 4.0 Gigabytes per second over a x16 PCI.

PCI Express architecture provides a high performance I/O infrastructure for Ethernet, TV Tuners, 1394a/b controllers, and general purpose I/O.

Photograph: Mery Xavier Imaging, Anil Deshmukh

The Magic Bus

Is PCI express the bus you have been waiting for? Time to get on it...

Varun Dubey

Man's need for speed is perhaps the strongest driving force behind his innovations; blazingly fast Ferraris and supersonic jet planes bear testament to this. Computing, of course, has not been left untouched. In fact, the advances in technology in terms of processing power have been so rapid over the past 10 years that had they been replicated elsewhere, we would have already visited the most distant galaxies!

One aspect of computing—the PCI bus—has, however, remained untouched for almost a decade! We demystify the latest buzzword in the tech world—PCI express.

Everyone On The PCI Bus!

We need to first understand how PCIe works, and why it is needed, let's first understand the basic working of a 'bus'. A bus is a bunch of conductors, and is primarily a means of transferring data from one system component to another. For example, a bus is used to transfer data from your hard disk to the printer, or from the processor to the memory, apart from other functions. PCI uses the shared bus architecture, where the devices attached to the bus share the bus' bandwidth when working.

It is interesting to note that the PCI bus is the oldest, slowest bus (or piece of technology, if you will) on your motherboard. Considering that it connects everything else—the latest video cards to the processors to the latest memory—it's a bit scary. In the infographic (*on page 33*), note how underserved devices connected to your USB or Firewire port or even your SATA-based hard disks are. This is the crux of the problem—the bandwidth requirement of a device is not satisfied by the PCI bus; the bus itself becomes the bottleneck that needs to be rectified for the solution.

For the second problem, consider a hard disk. Under the aegis of PCI, if this disk needs to communicate with the CPU, it will have to go via the Southbridge, stop at the Northbridge for refreshments and finally arrive at its destination, fighting latencies all along the way. "Latencies", you ask? Well, yes, notice how the USB, the FireWire and the Gigabit LAN ports are all sharing the same 133 MBps bus. Fighting for resources means system slowdown. This is problem number two in PCI-land.

The third problem with the PCI architecture is that it is a one-way lane. That is, when the CPU is talking to the Northbridge, no other device can access the CPU. This leads to a scramble for bandwidth and adds to system slowdown.

A slow bus fighting with fellow devices for resources, and clogging at the Northbridge, are